DIAGNOBOT

A Local AI-Assisted Clinical Triage Prototype

Skin image classification, clinical text support, and a proposed low-cost mobile health robot

Student author(s) _____	Department / college _____
Project guide _____	Academic year 2026-2027

College research paper draft. Complete author/institution details and verify dataset, hardware, and supervisor information before submission.

Prepared from project code, trained checkpoint, evaluation files, supplied prototype photograph, hardware inventory, and recent published research.

Abstract

DiagnoBot is a student prototype for local, image-assisted health triage. Its software combines a fine-tuned MobileNetV3-Small classifier for ten skin-image folder labels, Ollama Moondream for image descriptions, Ollama Llama 3.2 for short clinical-context summaries, and deterministic Python rules for symptom extraction and urgency cues. The wider codebase also includes a TorchXRayVision DenseNet-121 chest X-ray path. This paper describes the current software and proposed integration with a Raspberry Pi 5, ESP32, MAX30102 pulse-oximeter module, MLX90614 infrared thermometer, USB webcam, and motorized chassis. Sensor and motor integration is proposed; the repository contains no Pi GPIO or ESP32 control code. The classifier was trained on 932 images with a stratified image-level 70/15/15 split. On 140 held-out internal-test images it achieved 70.7% accuracy, 70.3% balanced accuracy, and 0.699 macro-F1. These results show technical feasibility only. Small test support, missing patient/source identifiers, undocumented dataset provenance, class confusion, and no external testing prevent clinical performance claims. The paper concludes with a plan for improved labels, grouped evaluation, focused scope, and safe hardware integration.

Keywords: MobileNetV3-Small; skin image classification; edge computing; health-monitoring prototype; ESP32; Raspberry Pi; evaluation metrics

Paper sections

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1. Introduction

Skin conditions are visually diverse. Similar presentations, lighting and camera variation, and differences in skin tone complicate image classification. Recent dermatology AI research ranges from compact classifiers to specialty foundation models trained on millions of images; provenance, label quality, and independent testing influence whether reported performance is credible [1-6]. DiagnoBot demonstrates a local decision-support pathway while making evaluation limits explicit.

DiagnoBot is a web prototype with skin-image classification, optional chest X-ray analysis, local vision/language model components, and deterministic triage rules. The planned physical robot adds a webcam and vital-sign sensors to a Raspberry Pi 5 and ESP32. Its purpose is to demonstrate sensing, edge processing, interface integration, and human review, not autonomous diagnosis.

The model roles are separate. MobileNetV3-Small predicts a skin-image class. Moondream produces image-description text. Llama 3.2 summarizes supplied context. Python rules extract terms and urgency cues. Classifier metrics do not measure generated text or the whole triage workflow.

1.1 Scope and contribution

This paper describes the software and hardware plan, reports the latest skin-classifier experiment, and proposes a safe route to integrate the robot. It does not claim a new clinical algorithm or validated medical device.

2. Problem Statement

The engineering problem is to build an affordable demonstration system that collects a skin image and basic vital-sign measurements, processes them locally, and presents a cautious summary for human review. The repository does not yet connect all data sources: it has no Pi/ESP32 sensor or motor-control implementation, and the MAX30102, MLX90614, and USB webcam were listed as planned purchases.

The ML subproblem is narrower: classify a skin image into one of ten folder labels. The dataset contains 932 images, with only 140 in the internal test partition. Dataset source, expert reference standard, and label adjudication are undocumented. Images were split at image level because patient identifiers were unavailable; related images may cross partitions. Performance on new patients, devices, or institutions is unknown.

Research question

Can a compact local classifier and rule-assisted interface demonstrate image-based triage, and what performance and integration limits appear in a small held-out image evaluation? This is a technical-feasibility question, not a clinical diagnostic-accuracy question.

3. Objectives

- Build a local web prototype that presents classifier output and clinical context for human review.
- Use MobileNetV3-Small for the ten skin-folder labels and report class-level errors as well as accuracy.
- Explain how Moondream and Llama 3.2 differ from the trained classifier and deterministic rules.
- Propose Raspberry Pi 5 and ESP32 integration for webcam, MAX30102, MLX90614, and chassis.
- Identify data, evaluation, and safety requirements before clinical claims.

4. Literature Survey - Recent Work

The 2024-2025 studies below cover few-shot learning, label noise, fairness, larger classifiers, clinical datasets, embedded monitoring, and reporting. Their results are not directly comparable to DiagnoBot.

[1] Chen, Liu, and Yang (2024) - CDD-Net and Derm104. The few-shot study introduced a 4,483-image, 104-category clinical skin dataset and a multiscale feature-fusion module. It reported gains up to 9.14 percentage points for selected few-shot settings. This is relevant to data scarcity, but category-holdout results differ from DiagnoBot's ten-class split.

[2] Shen et al. (2024) - contrastive learning and noisy labels. This study used online community images for contrastive pretraining and filtered noisy labels across 22 conditions. Reported top-1 accuracy moved from 42.05% to 45.05% after pretraining and reached 49.64% after filtering with a larger validation subset. It highlights label quality; internet imagery also creates population/source bias risks.

[3] Dip et al. (2024) - equitable skin prediction. The authors studied transfer learning and domain adaptation on the Diverse Dermatology Images dataset and emphasized performance gaps across skin tones. DiagnoBot has no skin-tone metadata or subgroup analysis, so diverse data and subgroup reporting are needed.

[4] Alruwaili and Mohamed (2025) - multi-architecture classification. Their EfficientNet-B0/B2 and ResNet50 ensemble used 27,153 Kaggle images and reported 99.14% accuracy on an 80/10/10 split. It is not a fair target for DiagnoBot: data, labels, and split differ, and random image splits may be optimistic if sources overlap. It shows why accuracy must be accompanied by data and split details.

[5] Yan et al. (2025) - PanDerm. This dermatology foundation model used over two million multimodal skin images from 11 institutions and four modalities, with broad benchmarks and reader studies. It is a much larger clinical research effort, not directly comparable to a compact student classifier, and demonstrates the value of diverse data and reader evaluation.

[6] Sultana et al. (2025) - SkinDisNet. This dataset paper describes 1,710 smartphone clinical images from Bangladeshi hospitals, six conditions, and curated clinical/demographic metadata. Its labels differ from DiagnoBot but illustrate the value of documented provenance.

[7] Pawar et al. (2024) - ESP32 health monitoring. This IEEE prototype uses ESP32, MAX30102, a temperature sensor, display, and connected app. It is relevant to embedded sensing but does not validate DiagnoBot or establish clinical-grade sensor readings.

[8] Tejani et al. (2024) - CLAIM 2024. The medical-imaging AI reporting checklist calls for transparent data sources, partition sizes/split levels, preprocessing, evaluation, and external testing. It distinguishes a held-out internal subset from a separate external dataset and generally recommends patient-level or higher disjoint partitions.

4.1 Research gap

A defensible student contribution is a transparent edge-computing prototype that separates classifier, sensors, rules, and language models; reports an honest internal test; and identifies evidence required for safe use. It is not a claim of a novel diagnostic algorithm.

5. Proposed Solution

The proposed system has two layers. The current software includes a FastAPI backend, dashboard, MobileNetV3-Small skin classifier, Moondream and Llama 3.2 model paths when model-backed mode is enabled, deterministic symptom extraction, and transparent triage rules. The broader codebase also has a TorchXRayVision DenseNet-121 route for chest X-rays. A proposed robot adds a webcam and sensor readings. The dashboard should show each output with its source, confidence, and limitations and require human review.

Model stack used in the project

Module	Model / implementation	Purpose and rationale
Skin	Torchvision MobileNetV3-Small; ImageNet default weights, fine-tuned in PyTorch	Compact transfer-learning classifier for CPU/edge experiments; ten outputs match local folder labels.
Image description	Moondream via local Ollama tag `moondream`	Natural-language image observations; separate from classifier, not benchmarked here.
Clinical text	Llama 3.2 via local Ollama tag `llama3.2`	Short summaries from supplied context; no project answer-quality benchmark.
X-ray	TorchXRayVision DenseNet-121, `densenet121-res224-mimic_ch`	Separate pretrained chest X-ray model; not evaluated in this skin experiment.
Triage	Deterministic Python rules	Inspectable keyword/range logic; not an LLM or clinically validated tool.

The code defaults to MOCK_MODE=true, so simulation/fallback behavior may be used instead of the Ollama models. Ollama identifiers are tags, not immutable model versions. Record model digests for repeatability. The X-ray path was not evaluated.

6. Working Principle

Figure 1 separates the current software flow from the dashed proposed hardware layer. An image is processed by the skin classifier and image-description model. Symptoms and vital signs are supplied through the application or, in the future, by sensors. Rules derive urgency cues. A language model may summarize context. The clinician-facing interface returns these signals for review. In the proposed robot, an ESP32 reads the sensor modules and sends timestamped data to the Pi; the Pi runs the backend and classifier and receives webcam images.

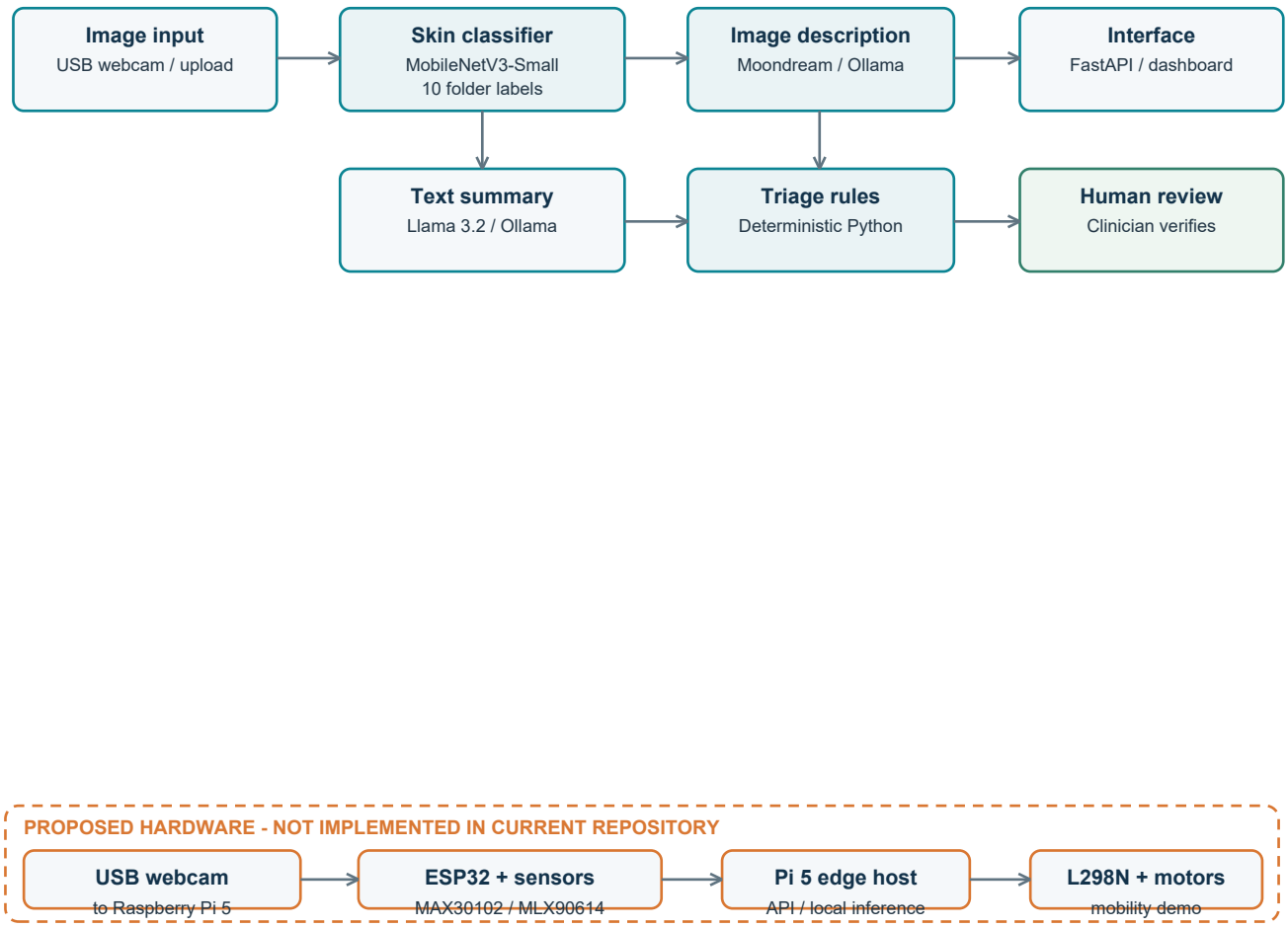


Figure 1. Current software workflow (solid boxes) and planned physical links (dashed orange). Hardware links are not implemented in this repository.

7. Methodology

7.1 Design

This is a retrospective engineering experiment on an existing folder-organized image collection with a descriptive review of the prototype. No prospective patient study, clinician reader study, or diagnostic workflow evaluation was conducted. The target is ten-way image classification using the folder names as labels.

7.2 Data and split

The dataset contains 932 images. A seeded (42), class-stratified image-level split assigned 652 to training, 140 to model tuning/validation, and 140 to internal testing. Patient and source identifiers were unavailable. The local files do not document dataset provider, licensing, labeler qualifications, or reference standard; these need verification before publication or redistribution.

Table 1. Partition counts

Class	Train	Tune	Test
Bullous	70	15	15
Eczema Photos	70	15	15
Lupus	70	15	15
Urticaria Hives	70	15	15
acanthosis nigricans	44	9	10
acne	70	15	15
candidiasis	70	15	15
impetigo	50	11	11
molluscum-contagiosum	68	15	14
nevus	70	15	15
Total	652	140	140

7.3 Preprocessing and training

Images were resized to 224 x 224, converted to tensors, and normalized with ImageNet channel means and standard deviations. Training-only augmentation used horizontal flips (p=0.5), rotations up to 10 degrees, and mild brightness/contrast/saturation jitter (0.12). Separate dataset instances kept augmentation out of tune/test. Batch size was 32.

The model used torchvision MobileNetV3-Small ImageNet default weights, with a replaced final layer for ten classes. It used class-weighted cross-entropy (weights proportional to the square root of inverse class frequency). The head was trained up to five epochs using AdamW (learning rate 5e-4; weight decay 1e-4), with features frozen. The final four feature blocks were then fine-tuned up to twelve epochs with AdamW (learning rate 2e-5; weight decay 1e-4). A plateau scheduler and four-epoch patience were configured. The run reached the 17-epoch cap; best validation macro-F1 was at epoch 15. The test set was evaluated once after selection.

The run used CPU and took 351 seconds. At epoch 15, training accuracy was 81.75% and tuning accuracy 67.14%, a 14.61 percentage-point gap. This suggests possible overfitting but is not a reliable future generalization estimate.

7.4 Proposed hardware procedure

Connect the MAX30102 and MLX90614 modules to ESP32 over their supported I2C/SMBus interfaces. Send timestamped readings to the Pi over USB serial as JSON. Connect the USB webcam to the Pi. Run the web application and classifier on the Pi after measuring memory and latency. Use the L298N with a separate motor supply; share signal ground where required, but keep motor current away from Pi/sensor rails. Stop the robot while measuring to reduce optical pulse-sensor motion artifacts.

8. Prototype (Bot)

Reported available hardware: Raspberry Pi 5, ESP32, L298N motor driver, chassis, DC motors/wheels, breadboard, and jumper wires. Planned purchases: MLX90614/GY-906 temperature sensor, MAX30102 heart-rate/SpO2 module, and USB webcam. The supplied photograph shows a Pi connected to a display with the DiagnoBot interface.

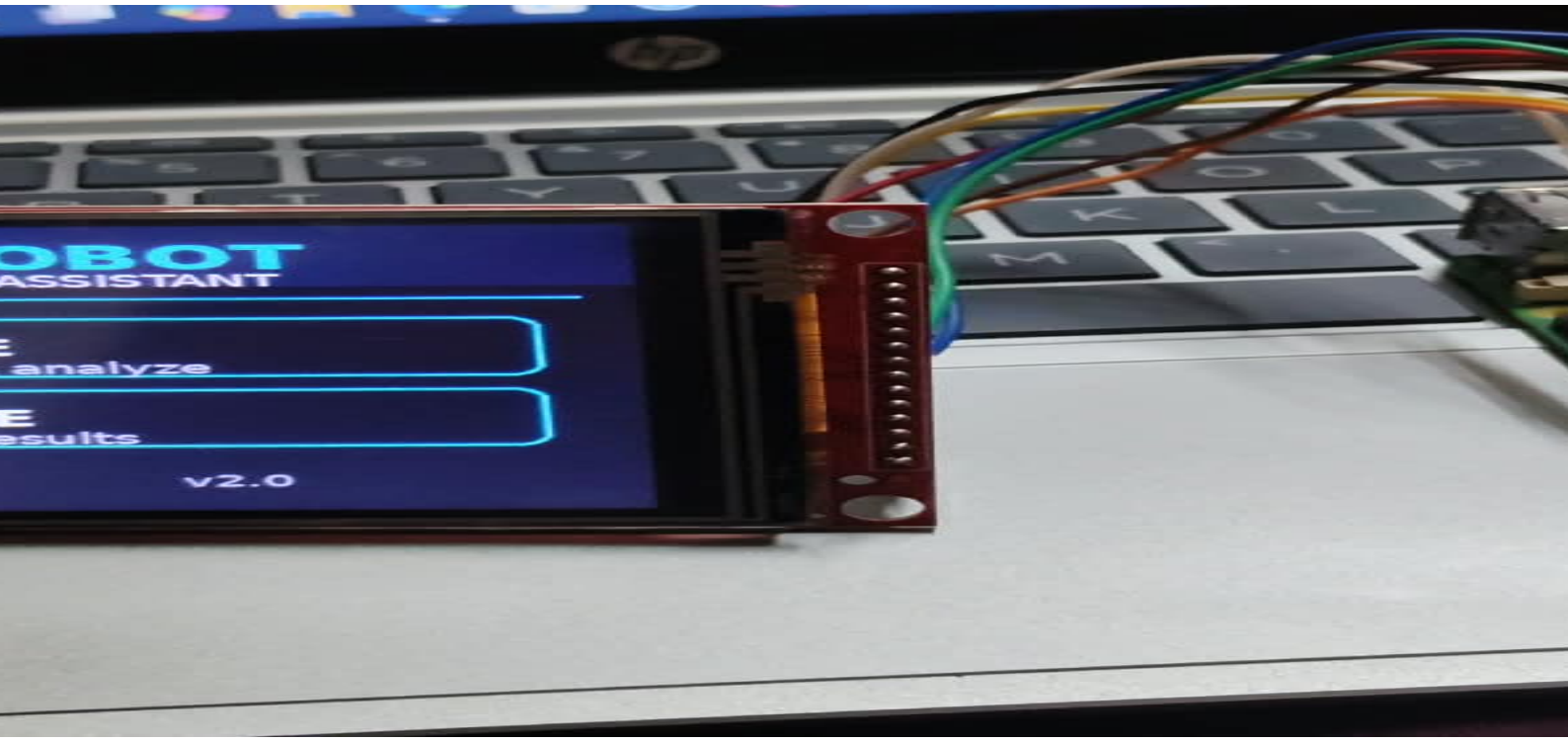


Figure 2. User-provided bench photograph of the Raspberry Pi and interface. It does not demonstrate sensor readings, motor control, or full integration.

8.1 Implementation status

Capability	Status
Skin classifier	Implemented; evaluated on small internal image split.
FastAPI/dashboard	Present in repository.
Moondream / Llama 3.2	Ollama paths exist when mock mode is disabled; answer quality is not evaluated.
DenseNet X-ray	Model-loading path exists; not evaluated here.
Pi/ESP32/sensors	No GPIO or serial control code in the repository; sensors listed as planned purchases.
Motors	Hardware listed; no motor-control code in repository.
End-to-end robot	Not evaluated.

8.2 Sensor and power considerations

MAX30102 is an optical heart-rate/pulse-oximeter sensor communicating over I2C. Its IC uses low-voltage supplies; breakout-board regulators and level shifting vary, so check the exact purchased module. MLX90614 is a non-contact infrared thermometer with SMBus output. It measures surface/object temperature and depends on target, distance, and environment; do not treat it as core temperature or clinical fever diagnosis without a validated procedure. Raspberry Pi GPIO uses 3.3 V logic. Use manufacturer documentation [9-12].

MAX30102 readings are sensitive to motion and contact. Stop and stabilize the robot before readings. Keep motor supply surges away from Pi/ESP32/sensor rails. Check L298N board limits and motor stall current. Timestamp readings and mark poor signal/contact so they can be excluded.

9. Evaluation Metrics and Results

9.1 Metric definitions

Accuracy is correct predictions divided by all images. For each class, precision = TP/(TP+FP), recall/sensitivity = TP/(TP+FN), and F1 = 2PR/(P+R). Macro-F1 is the unweighted mean of class F1 scores. Balanced accuracy is the mean class recall. Cross-entropy penalizes low probability assigned to the true class. A confusion matrix shows true versus predicted labels.

These values assess the image classifier only. They do not evaluate Moondream descriptions, Llama summaries, triage-rule safety, sensor accuracy, or the complete robot.

9.2 Per-class internal-test metrics

Class	n	Precision	Recall	F1
Bullous	15	0.700	0.467	0.560
Eczema Photos	15	0.467	0.467	0.467
Lupus	15	0.889	0.533	0.667
Urticaria Hives	15	0.588	0.667	0.625
acanthosis nigricans	10	0.875	0.700	0.778
acne	15	0.700	0.933	0.800
candidiasis	15	0.938	1.000	0.968
impetigo	11	0.600	0.545	0.571
molluscum-contagiosum	14	0.647	0.786	0.710
nevus	15	0.778	0.933	0.848

Table 2. Per-class test results (n is the number of test images).

9.3 Overall results and interpretation

Metric	Value	Meaning
Accuracy	70.7% (99/140)	Fraction of all test images correct.
Balanced accuracy	70.3%	Average recall across ten classes.
Macro precision	71.8%	Unweighted class precision.
Macro recall	70.3%	Unweighted class recall.
Macro-F1	0.699	Unweighted class F1.
Test loss	0.915	Cross-entropy on test predictions.
Approx. 95% Wilson CI	62.8%-77.7%	Image-level uncertainty interval; does not handle leakage/bias.

The Wilson interval is approximate for 99 correct predictions among 140 images. It does not account for repeated patients, possible source leakage, label noise, or camera/population shift.

The lowest F1 values were Eczema Photos (0.467), Bullous (0.560), and impetigo (0.571). Six of 15 Eczema Photos test images were predicted as Urticaria Hives. These errors suggest label review and more independent examples of visually overlapping categories before a model-architecture change. Candidiasis scored F1=0.968 on only 15 images; its estimate is also uncertain.

For a college paper, 70.7% can be reported as a preliminary internal experiment if sample size, labels, split, and limitations are explicit. It is not clinical accuracy and cannot be directly compared with studies using different datasets/protocols.

10. Discussion and Limitations

The experiment shows that a compact ImageNet-pretrained network can learn useful patterns from these ten folder labels. Overall accuracy, balanced accuracy, and macro-F1 are similar, but class-level performance varies. The internal test set is small; patient/source identifiers are absent; labels/provenance are unclear; and no external cohort was evaluated. Results may be optimistic and may not transfer to phone-camera images from the robot.

The updated training process separates training augmentation from tune/test transforms and uses an independent internal test partition. However, no split can compensate for weak labels, duplicated imagery, or poor representativeness. Data provenance, reference labels, and patient/source-grouped testing should be addressed before more architecture tuning.

The code includes generative vision and language components but no task-specific evaluation of their outputs. A future study should use a blinded clinician-reviewed rubric for factual observations, unsupported statements, missed findings, and consistency with reference notes. Triage rules need separate case-based validation. Sensor readings need comparison with suitable reference instruments; they must not be folded into skin-classifier accuracy.

Edge feasibility is also unmeasured. Pi 5 inference latency and RAM use should be profiled. Moondream/Llama inference may be slow or too large for the selected Pi memory/quantization. Switching to a hosted model would alter the privacy and connectivity assumptions. The current paper reports no Pi runtime benchmark.

10.1 Recommended next experiments

- Document dataset source, license, acquisition, definitions, label review, and expert/reference-standard process; detect exact and near duplicates.
- Use patient/source-grouped splits and keep a final external cohort untouched. If identifiers are unavailable, obtain a better-documented or newly collected dataset.
- Narrow the research task to one coherent condition/group with lookalikes and relevant negative examples; collect representative images across patients, skin tones, devices, lighting, and sites.
- Compare a majority-class baseline and one alternate pretrained model with identical grouped splits; tune only on training/tune data.
- Report class-wise sensitivity/specificity, precision, macro-F1, balanced accuracy, calibration, confidence intervals, and abstention behavior.
- Evaluate LLM/VLM, rules, sensor readings, and human workflow separately using predefined rubrics and reference measurements.
- Integrate hardware in stages; verify I2C/serial data, electrical isolation, latency, power, sensor repeatability, and human usability.

11. Conclusion and Future Work

DiagnoBot is a college engineering prototype combining a web application, compact skin-image classifier, local LLM/VLM paths, and deterministic triage code. The retrained MobileNetV3-Small scored 70.7% accuracy, 70.3% balanced accuracy, and 0.699 macro-F1 on an internal image-level test split of 140 examples. These are useful preliminary results but do not establish clinical accuracy. Class confusion, small class supports, undocumented provenance, possible image leakage, and no external test remain important limitations.

The physical plan uses the available Pi 5, ESP32, L298N, chassis, and motors, with planned webcam, MAX30102, and MLX90614 purchases. The repository has not implemented those links; this paper therefore describes them as proposed. Future work should prioritize data provenance, patient/source grouping, a focused task, external testing, Pi benchmarking, and staged sensor integration.

The strongest college-paper contribution is a transparent system design and evaluation, including measured results and limitations, rather than a claim that the model diagnoses skin disease reliably.

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Project artifacts consulted

Source code and run files: train_skin_mobilenet.py; ml_pipeline/vision/derm_cnn.py; ml_pipeline/vision/qwen_vl_analyzer.py; ml_pipeline/nlp/report_generator.py; ml_pipeline/vision/xray_analyzer.py; config/settings.py; evaluation/skin_test_metrics.json; evaluation/skin_test_confusion_matrix.csv. These are project evidence, not published literature.

Citation note: published-paper results describe those papers only. Do not rank them directly against DiagnoBot because datasets, tasks, split protocols, and reference labels differ.